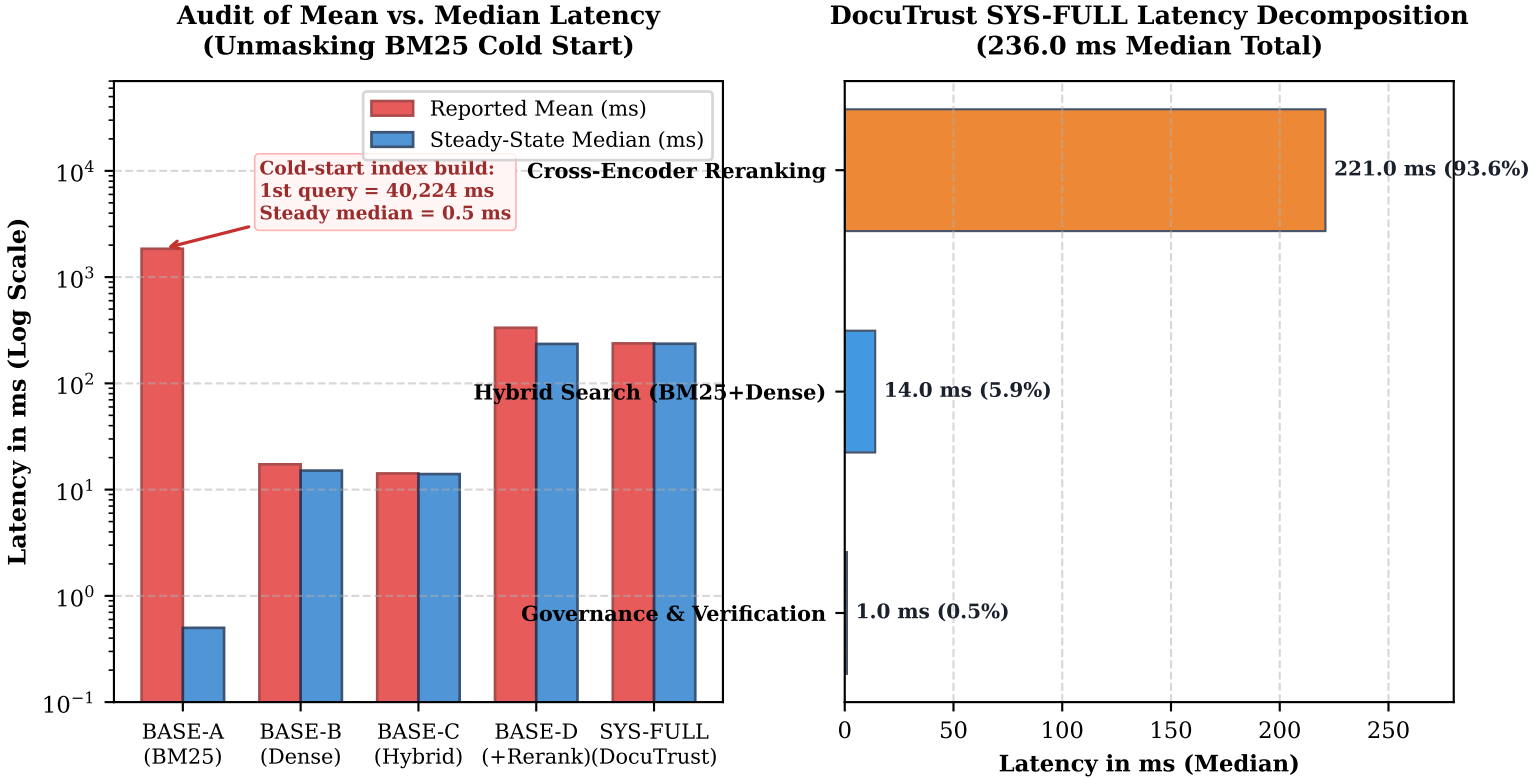
Audit of Mean vs. Median Latency
(Unmasking BM25 Cold Start)
Reported Mean (ms)
Steady-State Median (ms)
Cold-start index build:
1st query = 40,224 ms
Steady median = 0.5 ms
Latency in ms (Log Scale)
10^4
10^3
10^2
10^1
10^0
10^{-1}
BASE-A (BM25)
BASE-B (Dense)
BASE-C (Hybrid)
BASE-D (+Rerank)
SYS-FULL (DocuTrust)
DocuTrust SYS-FULL Latency Decomposition
(236.0 ms Median Total)
Cross-Encoder Reranking
221.0 ms (93.6%)
Hybrid Search (BM25+Dense)
14.0 ms (5.9%)
Governance & Verification
1.0 ms (0.5%)
0
50
100
150
200
250
Latency in ms (Median)